

SmartCare v0.3 - Domain Model Workbook

Week 6 student resource

Requirement-to-Concept Trace – Links to Week 5 Functional requirements

Requirement	Concept	State/behaviour	Decision
FR-01	Patient	Stores patient name	Accepted
FR-02	Patient	Searches for patient information	Accepted
FR-03	Practitioner	Stores practitioner information	Accepted
FR-04	Appointment	Creates appointment linked to patient and practitioner	Accepted
FR-05	Appointment	Stores appointment date and the time	Accepted
FR-06	Appointment	Displays Recorded appointments	Accepted
FR-08	Appointment	Updates appointment details	Accepted
FR-09	Appointment	Stores appointment status and cancellations	Accepted
FR-10	Appointment	History	Accepted

- FR-01: The system has to let staff add a patient record
- FR-02: Allow staff to view and search patient information
- FR-03: Allow staff to add and manage practitioner records
- FR-04: Allow staff to create an appointment and link that to the practitioner and patient
- FR-05: Record date and time for every appointment
- FR-06: System has to display recorded appointments
- FR-08: Allow appointment details to be updated
- FR-09: Allow appointment status to be recorded
- FR-10: Store appointment records so appointment history can still be viewed

CRC Cards

Patient

Responsibilities	Collaborators
Add patient information	Appointment
Find patient information	

Practitioner

Responsibilities	Collaborators
Add practitioner information	

Manage practitioner information	Appointment
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Appointment

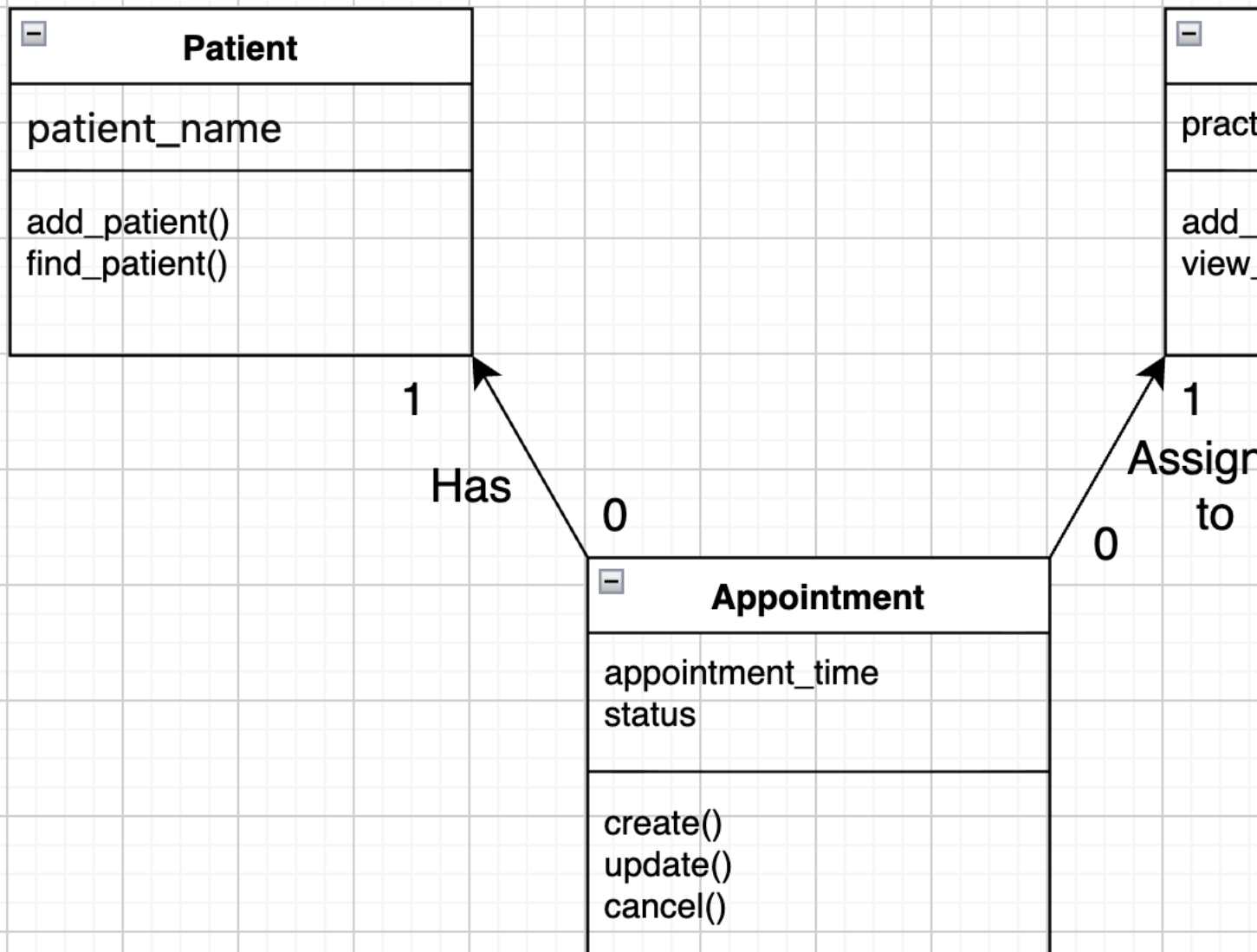
Responsibilities	Collaborators
Create, update and cancel	Patient
Store date and time and status and keep appointment history	Practitioner

Optional class

Responsibilities	Collaborators
Not needed. The requirements only cover the above three	

UML Class Diagram

Insert/draw UML here. Include defensible relationships and multiplicities.



Design Rationale

Explain class selection, responsibility allocation and key relationships.

Patient, Practitioner and Appointment are the main three classes as they are the main concepts supported and confirmed from the SmartCare Scope. Patient and Practitioner store the basic information needed to create an appointment. Appointment is the center class that links patient to a practitioner, and it stores the time, dates, details and the status.

The relationships is a patient can have none or many appointments. A practitioner can have none or many appointments. Each of the appointments are linked to one patient and one practitioner

AI Design Review Record

AI suggestion	Evidence	Decision	Reason	Model change
PatientManager	FR-01,02	Rejected	Do not require a separate class	No Change
PractitionerManager	FR-03	Rejected	Not required	No Change
SppointmentManager	FR-04,06,08,09,10	Changed	Appointment management is required but to keep it simple it can be placed in Appointment rather than adding another class	Appointment is the core appointment
ClientController	FR-01,10	Rejected	Not a confirmed requirement	No Change
NotificationManager	None	Rejected	Not a confirmed requirement	Change